

Portfolio of Burcu Erarslanoglu

Machine Learning • Signal Processing • Medical Imaging

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Gaussian Process Regression on a Grid with Tensor Train Matrices

Sep 2024 - Nov 2024

Gaussian Process Regression (GPR) was implemented for large grid-structured datasets using tensor train representations to exploit Kronecker structure in covariance matrices. The approach significantly reduced the computational complexity of GPR while preserving prediction accuracy.

Hand Gesture Classification Based on EMG Signals

Sep 2024 - Nov 2024

A machine learning pipeline was developed for classifying hand gestures using electromyography (EMG) signals collected from multiple subjects. Feature extraction was performed in both time and frequency domains, and classifiers including KNN, SVM, and MLP were evaluated using cross-validation. Model performance was optimized through hyperparameter tuning and comparative analysis.

Methods to Derive FC with fMRI Using BSS with Tensor Decomposition

Apr 2024 - Aug 2024

Advanced methods such as blind source separation and tensor decomposition were reviewed for extracting functional connectivity of the brain from fMRI data. The study provided comparative analyses of these methods and discussed their applications in both medical fields, addressing key challenges in high-dimensional neuroimaging data.

Linear Discriminant Analysis for Recognising Differences

Apr 2024 - Jun 2024

PCA-LDA framework was applied to MRI brain scans to classify Alzheimer's disease by reducing the data's dimensionality for improved classification. The approach demonstrated potential for early diagnosis of neurodegenerative diseases through k-fold cross-validation, optimizing classification accuracy across different principal components.

Array Parameter Estimation and Speech Enhancement for Array Processing

Apr 2024 - Jun 2024

Studied methods used in the estimation of directions and frequencies in an array and channel equalization. Compared the performances of EVD and GEVD on the three different beamformers by using SNR and STOI metrics.

Wavefield Imaging on 2D Linearized Inverse Scattering Problems

Apr 2024 - Jun 2024

Two-dimensional linearized inverse scattering problem was solved to obtain an image of an object with respect to a contrast function.

Boolean Compressed Sensing

Feb 2024 - May 2024

The problem of boolean compressed sensing in regards to group testing was explored using several algorithms to effectively recover sparse measurements.

Compressed Sensing and Machine Learning Techniques in MRI

Feb 2024 - May 2024

This paper explored the approaches of compressed sensing and machine learning in MRI. For these approaches, examples of various techniques and applications were given and their potential to enhance imaging efficiency and quality were discussed.

Towards Reconstructing the Brain with Graph Signal Processing

Feb 2024 - Apr 2024

GSP was applied to understand the structural and functional connectivity of the brain, as well as diagnosing mental disorders. The study explored the potential of combining GSP with deep learning for enhanced neuroimaging analysis and connectivity matrix generation.

Compressed Sensing for Applied Convex Optimisation

Nov 2023 - Jan 2024

A compressed sensing-based signal reconstruction as a suitable convex optimization problem was formulated and then the signal reconstruction algorithm was implemented. The signal was reconstructed using off-the-shelf solvers and a low-complexity algorithm of projected subgradient decent.

Accelerometer Calibration for Estimation and Detection

Nov 2023 - Jan 2024

Various estimators like the Maximum Likelihood Estimator (MLE) and performance bounds like the Cramer-Rao Lower Bound (CRLB) were applied for the challenge of calibrating the accelerometer.

Fusing Noisy and Corrupted MRI Images*Sep 2023 - Nov 2023*

The project addresses the challenge of improving MRI image quality in scans of the eye. The outliers that happen in MRI scans due to factors like patient motion and noise are detected and restored by optimal filtering and spectral estimation. Different filters are used to improve the image quality and results are compared using SNR.

Online Music Platform Database Application*Feb 2023 - June 2023*

Designed and implemented a database-backed online music platform supporting multiple user roles. The system enabled operations such as track browsing/filtering, album/playlist management, user history tracking, and social interactions including following users and liking playlists. The project involved designing relational database schemas and implementing queries for managing music metadata, user activity, and artist collaborations.

Industrial Design Project: Detection of Forest Fires with WSNs*Sep 2022 - July 2023*

This project is carried out with METEKSAN Defence on the scope of EEE 493/494 which is a capstone design course. By considering the placement of sensors, power efficiency, communication protocols and terrain propagation effects; algorithms will be proposed and tested with MATLAB simulations for the detection of forest fires with WSNs.

Medical Imaging Project: Brain MRI Simulator*Dec 2022 - Jan 2023*

In this project, main contrast mechanisms of MRI namely T_1 , T_2 , PD weighting, flip angle, Gaussian and spike noise effects were studied by following three stages of MRI, which are slice selection, k-space acquisition and image reconstruction. [Video Link](#).

CMOS VLSI Design Project: SRAM Design*May 2022 - Jun 2022*

In this project, an 8x4 memory array and an 8x4 SRAM array was designed and combined in Cadence Virtuoso. Testbench results were obtained and power optimization was done. [Video Link](#).

Electronic Circuit Project: Design of D and T Flip-Flop*Sep 2021 - Dec 2021*

The design of D and T Flip-Flop was done by using BJTs. The proposed design was simulated on LTspice and tested on a breadboard. The proposed design was implemented on a PCB which was designed using DipTrace.

Microprocessors Project: Smart Car*Feb 2021 - May 2021*

In this project, FRDM-KL25Z board was used and the coding was done in C++ through the Mbed Compiler. The main components of the project were TCS3200 color sensor, HC-SR04 ultrasonic proximity sensor, L298N dual full-bridge driver, 4x4 keypad and 8x8 dot matrix board. [Video Link](#).

Digital Circuit Design Term Project: Weather Station*Sep 2020 - Dec 2020*

In this project, BASYS3 board was used and the coding was done in VHDL. The main components of the project were DHT22 sensor, LPS25HB sensor, 16x2 LCD display and a VGA monitor. [Video Link](#).